Frozen tangent predictions across every declared ridge penalty

24 checkpoints; no weight update. Each ratio uses that checkpoint and split's original MSE. Dotted line: no correction.

Controller
RMS-equalized counterfactual
P
Full FFN
Calibration anchors
Exposed evaluation anchors

Scalar mul | Scalar add | Vector mul | Vector add | SwiGLU 96 | SwiGLU 128

Ring8 · step 1000
Residual MSE / original MSE

Ring8 · step 5000
Residual MSE / original MSE

Product16 · step 1000
Residual MSE / original MSE

Product16 · step 5000
Residual MSE / original MSE

10
5
2
1
.5
.1
.05
0

1e−4
.01
1
10

Fixed ridge penalty λ

Y scale: linear below .05, logarithmic above. Full FFN is the only SwiGLU component. Stage15 cross-family FiLM initialization was not paired.